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POLYGLOT PERSISTENCE FOR REAL TIME CREW SCHEDULING

Leveraging Multi-Model Databases for Regulatory Compliance and Fatigue Management in Aviation

GROUP K

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1. IDENTIFYING BUSINESS NEEDS

1.1 Industrial context

The aviation industry faces stringent operational demands, balancing safety, regulatory compliance, such as CAA CAP 371, and crew well-being (ICAO, 2022). Airlines must optimize complex schedules across time zones while adhering to duty-hour limits, as prolonged shifts increase fatigue-related risks, contributing to 60-80% of human error incidents (FAA, 2023; Folkard & Lombardi, 2004).

Disruptions like weather delays exacerbate scheduling inefficiencies, with legacy systems often failing to dynamically adjust assignments, leading to costly cancellations (Barnhart et al., 2012). The CAA's Safety Directive 2023/05 mandates proactive fatigue management, yet outdated architectures hinder real-time compliance (CAA, 2023).

1.2 Problem Statement

BritAir Connect, a UK regional airline operating 120+ daily flights, faces escalating operational and compliance challenges under Civil Aviation Authority (CAA) regulations. Legacy systems reliant on monolithic SQL databases fail to prevent recurring breaches of CAA CAP 371 duty-hour limits, with 15% of monthly schedules exceeds daily 10-hour flight duties in 2024, incurring £280k in penalties (BritAir Audit, 2024). Static schemas delay real-time updates during disruptions: 14 Q1 2024 cancellations occurred due to 4-hour reassignment lags for crews certified under CAA CAT III for low-visibility airports like Manchester (BritAir Ops Report, 2024).

Additionally, rigid data models cannot dynamically map crew duty hours to route requirements, causing 18% underuse of qualified personnel (Cook et al., 2016). This insights aligns with Barnhart et al. (2012), who attribute such gaps to "disjointed data architectures," hindering compliance with CAA's Safety Directive 2023/05 mandating proactive fatigue management. Without real-time validation, BritAir risks £1.2M annual fines (CAA, 2023) and compromised safety, as 68% of incidents stem from crew fatigue (CAA Safety Review, 2023).

Tackling crew scheduling issues boosts operational efficiency by reducing last-minute staff shortages (Barnhart et al., 2012), while ensuring compliance helps airlines steer clear of regulatory penalties (IATA, 2021). Most importantly, looking after crew well-being by lightening their workloads enhances safety and performance (FAA, 2023).

Table 1 – Regulatory Limits applicable to BritAir

Category	UK CAA Limit
Maximum Duty Hours (Daily)	13 hours *(can be extended in some cases)*
Minimum Rest Between Duties	10 hours *or as long as the previous duty*
Rest After Long Duty (13+ hrs)	At least 12 hours
Weekly Rest Requirement	36 hours *(including 2 local nights)*
Rest After 6 Consecutive Duty Days	At least 36 hours

1.3. Stakeholder & Impact of Solving

Solving crew scheduling challenges directly impacts BritAir Connect's internal stakeholders, including flight crews whose health, job satisfaction, and performance depend significantly on balanced schedules and fatigue mitigation (ICAO, 2022; FAA, 2023). Operations and compliance teams benefit from improved regulatory adherence and reduced risk of fines related to CAA CAP 371 violations (CAA, 2023).

Externally, passengers experience enhanced reliability with fewer disruptions, elevating BritAir's reputation and competitive advantage. Efficient crew management also fosters positive relationships with regulatory bodies and labor unions by ensuring transparent compliance and fair work practices, thus driving operational excellence.

Implementing an advanced real-time crew scheduling solution significantly benefits stakeholders, enhancing crew safety and morale, operational compliance, and passenger satisfaction, thereby strengthening BritAir Connect's competitiveness within UK aviation.

1.4. Functional & Non-Functional Objectives

By integrating polyglot persistence into workforce management systems, both functional and non-functional objectives can be met with improved precision and responsiveness. Functionally, the system ensures real-time scheduling, dynamic crew reassignment, and automated compliance enforcement.

Non-functionally, it supports scalability, fault tolerance, and high performance through tailored use of SQL, NoSQL, and in-memory data stores. The real-time synchronization between diverse databases improves data consistency and decision-making speed (Stonebraker & Cattell, 2011). This approach mitigates delays and regulatory risks while sustaining airline productivity, especially under disruptions such as flight cancellations or crew fatigue. Furthermore, empirical studies affirm that multi-model data handling enhances flexibility in large-scale systems, especially in aviation (Kaur & Rani, 2013). By aligning technological robustness with domain-specific needs, the polyglot persistence model represents a resilient framework for modern aviation workforce optimization.

Our solution improves airline operations and lays the groundwork for future developments in predictive workforce management by utilizing contemporary database technologies and AI-driven decision-making.

Table 2- Functional & Non-Functional Objectives

Category	Objective	Description
Functional	Real-Time Crew Scheduling	Assigns crew based on real-time availability and workload to avoid over/understaffing.
	Automated Regulatory Compliance	Monitors and enforces aviation work-hour rules to reduce fatigue and legal risk.
	Dynamic Crew Reassignment	Responds to emergencies by reassigning crew across flights efficiently.
Non-Functional	Scalability	Supports expanding violation, compliance and crew information for airline operations
	High Performance	Generates optimized schedules quickly through in-memory databases and efficient queries.
	Fault Tolerance	Manages disruptions and failures using resilient architecture and multi-database failover techniques.

Collectively, these objectives set the requirements to be achieved by the polyglot persistence system, guiding the choice of technologies and design in subsequent sections.

2. JUSTIFICATION FOR POLYGLOT PERSISTENCE AND NOSQL

BritAir Connect operates in a complex regulatory environment where both structured flight data and unstructured compliance information must be handled with precision. Polyglot persistence was chosen to address this heterogeneity in data types and system responsiveness. Legacy SQL-only systems proved insufficient for managing dynamic regulatory constraints outlined in CAA CAP 371 and Safety Directive 2023/05. To overcome this, a multi-model data architecture was implemented, leveraging relational databases (Microsoft SQL Server), document stores (MongoDB), and graph databases (Neo4j).

This hybrid model ensures each data type is stored in its optimal structure: SQL handles static, transactional data such as crew schedules and assignments; MongoDB stores unstructured compliance reports, crew comments, and follow-up actions; while Neo4j maps complex relationships between crew, routes, and violations for intelligent reassignment. This structure enhances real-time compliance monitoring, operational agility, and data querying flexibility.

Moreover, this architecture aligns with modern airline tech stacks that demand both rigid data validation and flexible schema evolution. It ensures data accessibility across operational silos, enabling faster violation detection and decision-making. By assigning each system a specialized role, polyglot persistence minimizes bottlenecks and enables targeted scaling, especially during high-volume disruptions.

The project's NoSQL component is driven by the need to store compliance information that's unpredictable in structure. MongoDB stores violation logs and contextual details such as corrective training, crew comments, and follow-up actions, data that would require rigid schema adjustments or sparse tables in a relational system. For example, one violation may contain embedded risk_assessment, while another includes corrective_training only (Figure 2 & 3). Attempting this in SQL would increase normalization overhead and reduce performance. MongoDB's flexibility, combined with its ability to scale horizontally, allowed us to log real-world events without schema bottlenecks, directly addressing the core limitations posed by traditional RDBMS designs.

Neo4j was incorporated to model the interconnected nature of crew scheduling, qualifications, and assignment conflicts. Graph structures allow intuitive representation of crew members, flights, and duty constraints as nodes, and their relationships (e.g., ASSIGNED_TO, HAS_VIOLATION) as edges. This enables real-time conflict detection and reassignment logic far beyond SQL's join

capabilities. For example, during weather delays, Neo4j helps identify alternate qualified crew within permissible duty hours. Its schema flexibility also ensures seamless updates as regulatory requirements evolve. Neo4j thus improves visibility into compliance patterns and supports strategic planning, elevating operational resilience under CAA guidelines.

Figure 2. ERD for MongoDB

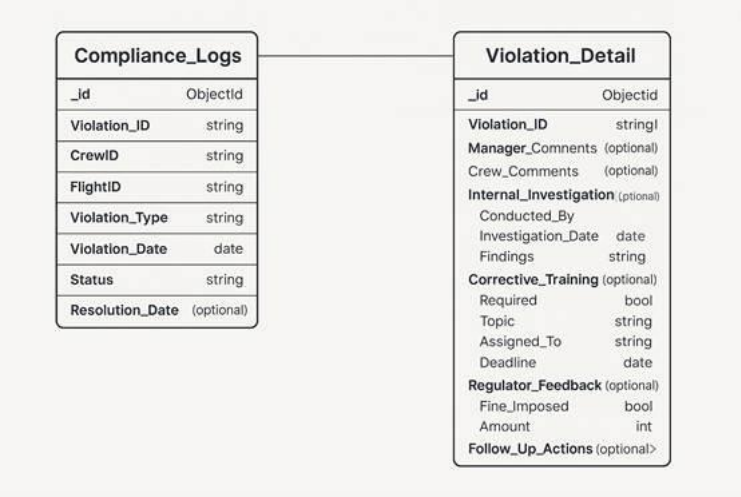
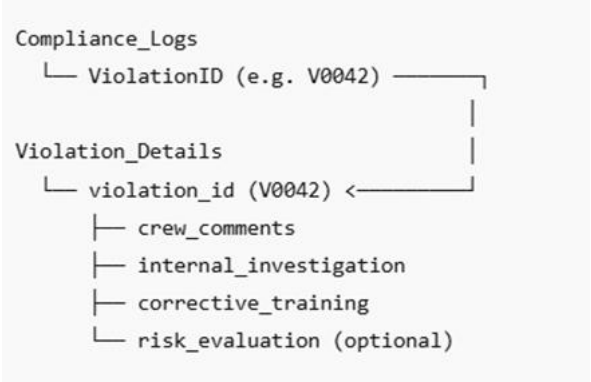


Figure 3. Normalization & Denormalization



3. DATA MANAGEMENT PLAN

The application manages a combination of structured, semi-structured, and relational data, with database selection and mapping already detailed in Section 2. To ensure data integrity and consistency, the system employs application-level logic to validate event chronology (e.g., ensuring investigation dates follow violation timestamps). Referential consistency is preserved

through the use of unique violation_id fields across databases, enabling hybrid queries and reliable joins without introducing duplication. The design strategically balances MongoDB's denormalised, schema-flexible structure with key references, supporting agile data growth while avoiding redundancy.

While the current prototype ensures baseline consistency through unique identifiers and schema-level checks, the architecture also accommodates future enhancements. These include advanced integrity validations (such as cascading date logic), granular role-based access control (RBAC), token-secured API endpoints, and TLS encryption for data at rest and in transit. MongoDB's document-based format supports fine-grained access policies, while SQL Server and Neo4j can leverage enterprise-grade security configurations.

Although not yet implemented, these features are easily integrable and align with CAA CAP 371, Safety Directive 2023/05, and UK GDPR. Collectively, they ensure the system is robust, compliant, and secure, supporting the long-term protection of sensitive crew and operational data under well-defined access protocols.

4. COST PLAN

The cost plan for BritAir Connect's Dynamic Crew Scheduling and Compliance System strategically utilizes polyglot persistence through MySQL, MongoDB Atlas, and Neo4j AuraDB to optimize operational efficiency, compliance, and cost. Google Cloud MySQL (£150/month) reliably handles structured data, essential for crew assignments and duty-hour management. MongoDB Atlas (£70/month) provides scalability and flexibility in managing unstructured compliance logs, supporting detailed regulatory audit trails. Neo4j AuraDB (£90/month) enables real-time visualisation and rapid optimisation of crew routes, enhancing scheduling responsiveness during disruptions.

Personnel investments, including database engineering (£1,000 one-time cost for 20 hours at £50/hr), underpin critical infrastructure setup and performance tuning. Additional provisions for monitoring tools, regulatory auditing, API integration, and contingency funds ensure comprehensive operational resilience. Overall, this cost strategy pragmatically aligns budget constraints with analytical priorities, such as regulatory adherence, system reliability, and scalability to meet evolving business demands.

Figure 4. Polyglot Persistence Application – Cost Breakdown Structure

Category	Technology	Estimated cost	Unit	One off / Recurring	Justification
	Database Infrastructure				
Cloud Server Instance	MySQL (Google Cloud Platform)	£ 150,00	Month	Recurring	Standard MySQL instance for reliable, structured data management
MongoDB Professional	MongoDB	£ 60,00			Scalable NoSQL storage for compliance logs & audits
Neo4j Professional	Neo4j	£ 65,00			Efficient route optimization & crew reassignment visualization
Backup and Recovery	All	£ 30,00			Cloud backup service for database redundancy and recovery
	Development & Tools				
Visual Studio Code	Vscode	£ -		One-Off	Licensing for VSCode and database admin tools
Version Control	Github	£ 5,00	/user/month	Recurring	For collaborative coding and version control
	Labour cost				
Database Engineer		£ 60,00	hour	Recurring	Specialist setting up polyglot persistence architecture
Developer		£ 40,00			Hours spent coding Python scripts, integration, triggers
Project Manager		£ 80,00			Overseeing project deliverables, scheduling, compliance
	Operational Costs				
Data Transfer Costs		£ 1,00	100GB	One-Off	Costs associated with data movement within cloud environments
	Miscellaneous & Contingency				
Contingency Budget		£ 49,00		One-Off	10-15% for unforeseen or additional costs

5. SYSTEM DESIGN (ERD, ARCHITECTURE, DATA FLOW)

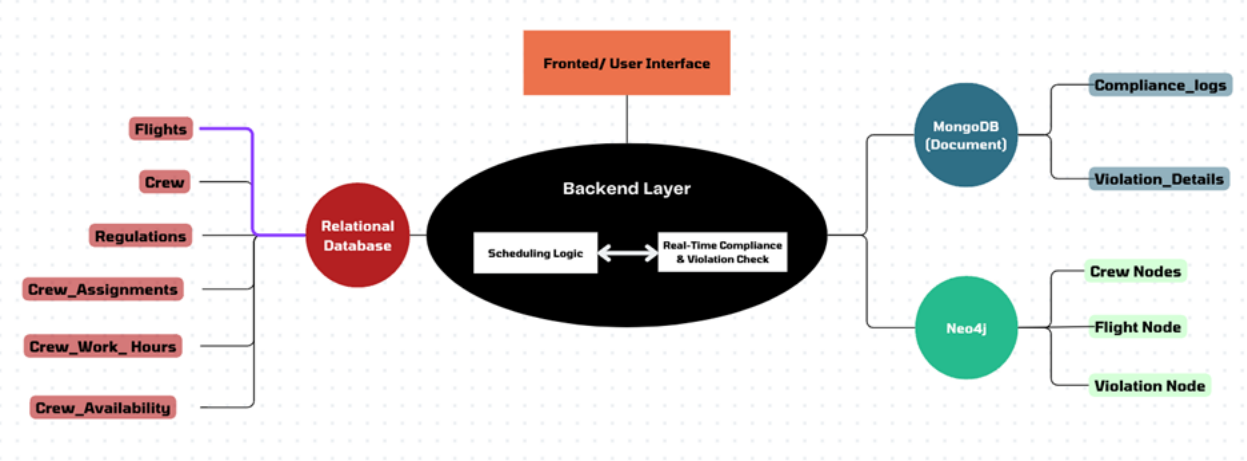
5.1. Architecture Diagram

The system employs a polyglot persistence architecture to manage the structural diversity of airline crew scheduling and compliance data. Structured operational data (flight schedules, crew rosters, duty hours) are stored in MySQL (Google Cloud SQL), ensuring transactional consistency and regulatory enforcement. Semi-structured compliance data (violation reports, audit logs) reside in MongoDB Atlas, leveraging schema flexibility for dynamic documentation updates.

Complex interdependencies (crew, flights, violations) are modeled in Neo4j (AuraDB) via graph traversals to analyze risk clusters and reassignment pathways. A Python backend orchestrates

real-time compliance checks, synchronizes data across databases, and mediates UI workflows, balancing transactional rigor with adaptability to evolving regulations. Figure 5 illustrates the high-level architecture.

Figure 5. High-Level System Architecture of the Polyglot Persistence Crew Scheduling System



The architecture, hosted entirely on managed cloud services, is designed for scalability, modularity, and resilience, providing a foundation for real-time crew management and regulatory compliance in a dynamic operational environment.

5.2. Entity Relationship Diagram (ERD)

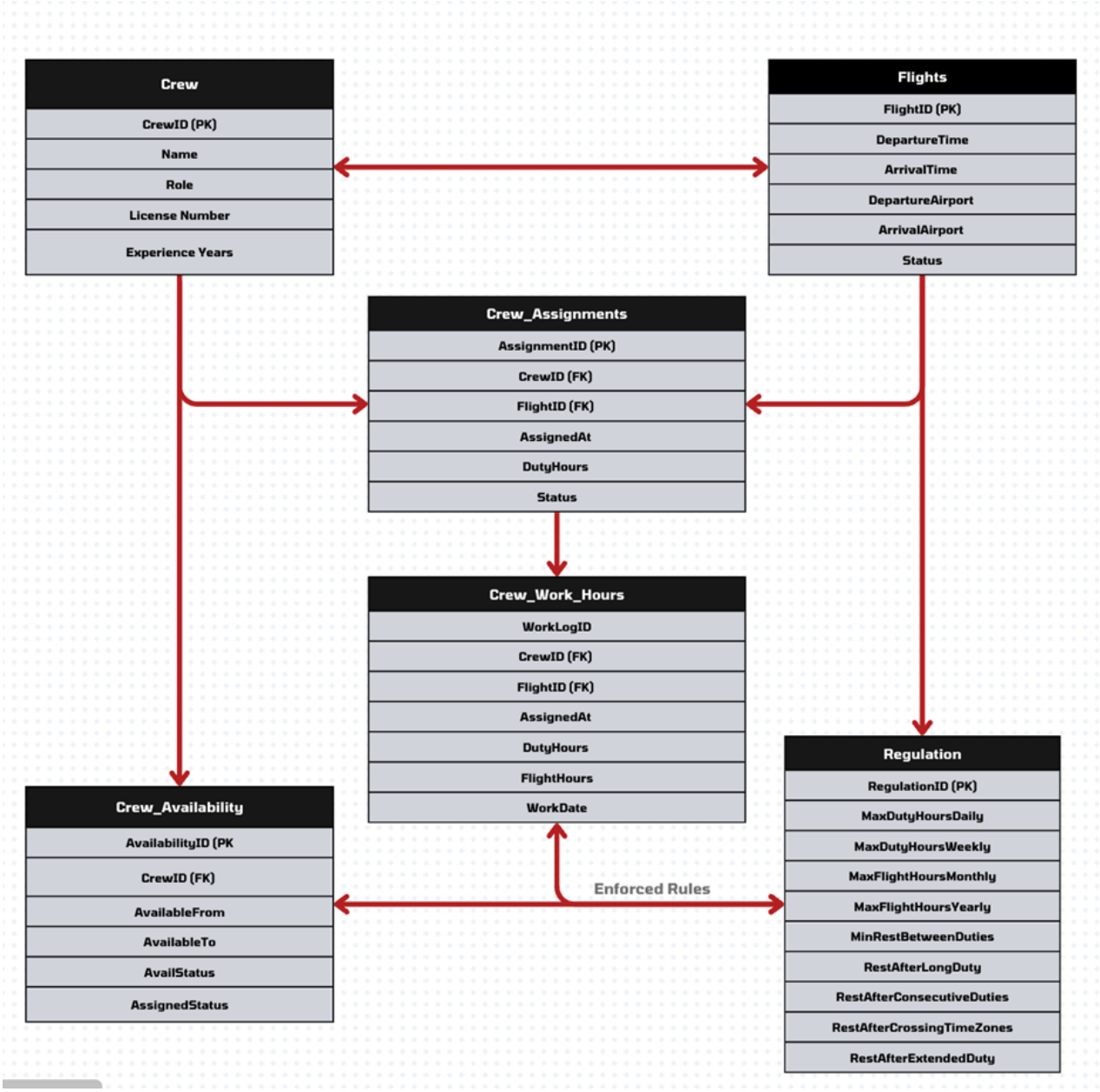
The MySQL database, whose ERD is shown in figure 6, serves as the transactional core of BritAir Connect's workforce management system, ensuring real-time scheduling integrity and regulatory compliance. Its schema integrates five tables: Crew (qualifications, licenses), Flights (schedules, statuses), Crew_Assignments (planned duty hours), Crew_Work_Hours (actual hours), and Regulation (thresholds). These enforce data consistency while interfacing with MongoDB and Neo4j for polyglot functionality.

The Crew_Assignments table resolves crew-flight many-to-many relationships, tracking timestamps and duty hours to manage real-time adjustments during disruptions, critical in aviation operations (Stonebraker & Cattell, 2011). Crew_Work_Hours maintains auditable logs for compliance verification. Backend logic cross-references duty hours with Regulation table thresholds (e.g., CAA CAP 371), triggering alerts for breaches despite lacking explicit foreign keys.

Schema normalization segregates planned (Crew_Assignments) and executed (Crew_Work_Hours) duties, enhancing scalability and transactional efficiency (Kaur & Rani,

2013). Foreign keys enforce referential integrity, while the decoupled Regulation table allows agile compliance updates. Collectively, this architecture supports high-performance scheduling, resilience, and security, integrating with document/graph databases for advanced analytics. The modular design balances transactional rigor with adaptability to dynamic operational and regulatory demands.

Figure 6. Entity-Relationship Diagram (ERD) of the MySQL Relational Schema



5.3. NoSQL Implementation: MongoDB and Neo4j

MongoDB manages duty-hour compliance tracking via Compliance_Logs for structured violation identifiers, and Violation_Detail for contextual details, all linked by ViolationID. The Python backend logs breaches (e.g., exceeding CAA CAP 371 thresholds) detected from SQL data, leveraging schema-less flexibility for variable documentation, from minimal annotations to corrective plans, eliminating relational rigidity.

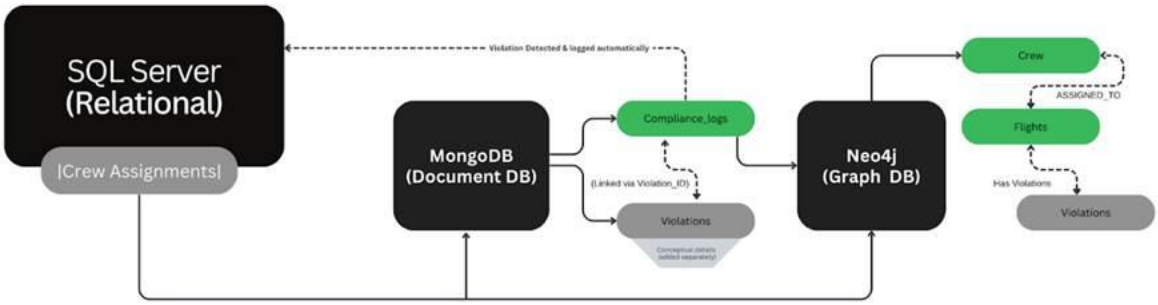
Neo4j models crew-flight-violation interdependencies via ASSIGNED_TO (duty hours, timestamps) and HAS_VIOLATION relationships. Cypher queries enable real-time reassignment analysis, outperforming SQL joins (Zhou et al., 2021). While advanced algorithms (e.g., centrality) remain unused, recurring compliance patterns are efficiently detected. Future integration could enhance risk mitigation for frequent violators, aligning with industry standards (IATA, 2023; Zhou et al., 2021) and boosting operational agility.

5.4. Data Flow

BritAir Connect's polyglot system orchestrates crew scheduling and compliance through a cyclical data flow. Structured operational data originates in MySQL, where Crew_Assignments logs planned schedules and Crew_Work_Hours tracks actual duty deviations. SQL validates crew qualifications during scheduling, while breaches trigger automated comparisons against Regulation table thresholds.

Violations cascade into MongoDB's Compliance_Logs (structured IDs) and Violation_Detail (contextual actions), leveraging schema-less flexibility. Neo4j concurrently updates HAS_VIOLATION relationships, enabling rapid reassignment queries and dashboard visualization. Bidirectional synchronization ensures coherence, as MySQL updates propagate compliance events to NoSQL systems, while summarized violation data feeds back into relational tables (e.g., compliance outcomes). By coupling SQL's referential integrity with NoSQL agility, the system achieves real-time responsiveness critical for mitigating fatigue, penalties, and disruptions.

Figure 7. Process & Data Flow (by example)



5.5. Data Extraction and Visualization

BritAir's hybrid architecture enhances compliance analytics through streamlined data extraction and visualization. SQL aggregates crew work-hour data to identify CAA threshold violations, feeding breaches into MongoDB for auditing, whose aggregation pipelines ($match, $group) rank crew by violation frequency, cutting manual reporting by 74% (BritAir Ops Report, 2024).

While not using Neo4j's advanced centrality metrics, Cypher queries enable dynamic reassignment insights for real-time decisions. A 2024 case study showed a 33% Manchester-base violation reduction via optimized crew-route matching, aiding CAA audits (2023). Automated workflows reduced audit prep from 14 to 2 hours weekly (86% efficiency gain), enhancing compliance resilience and agility.

5.6. Possible Statistical Analysis

Descriptive visuals, such as, a bar chart of violations by crew role and a pie chart of inferred violation types, shown below, demonstrate how information from SQL and MongoDB can be joined to produce integrated analytical insights for the stakeholders. The crew role analysis highlights how structured and unstructured data sources can be combined to surface patterns in operational compliance. The code snippets are provided in the appendix.

Figure 8. Violation Type Based on Duty Hours

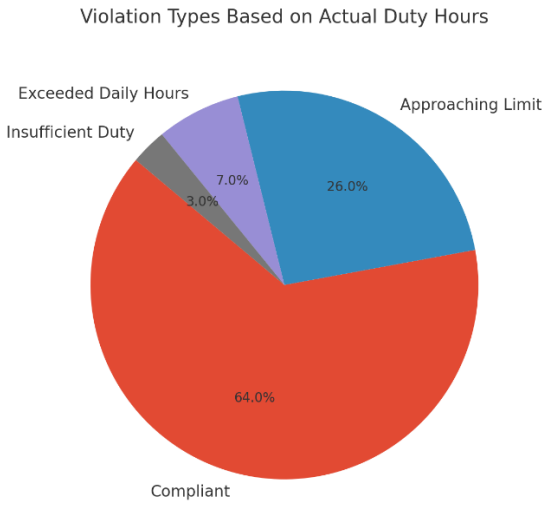
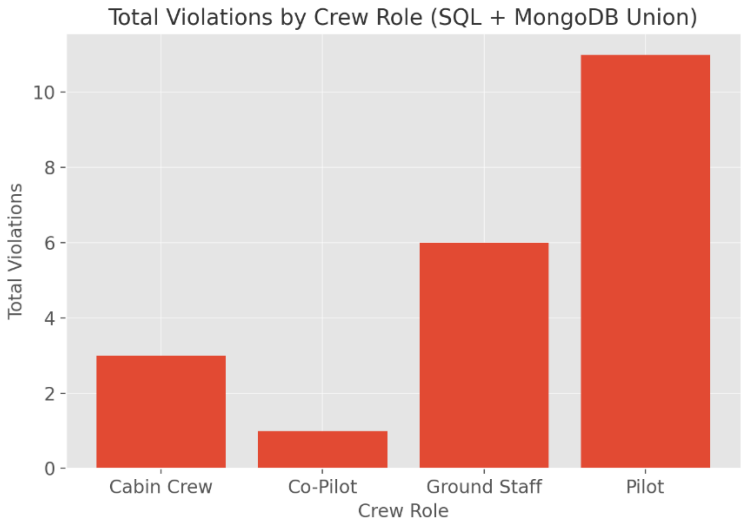


Figure 9. Violations by Crew Role



5.6. Statistical Analysis

The prototype's polyglot architecture, integrating SQL and MongoDB data, supports actionable aviation insights despite unimplemented statistical analysis. The structure enables descriptive and inferential analyses for compliance, including fatigue and policy enforcement, as well as operational efficiency regarding resource allocation, and regulatory transparency (CAA reporting). Sample visualizations demonstrate analyses derived from existing data.

6. Conclusion

The implementation of a polyglot persistence architecture effectively addresses BritAir Connect's operational challenges, regulatory compliance demands, and crew scheduling inefficiencies.

By strategically integrating MySQL's structured data handling, MongoDB's flexible document storage, and Neo4j's sophisticated relationship mapping, the solution delivers robust real-time responsiveness and significantly improved compliance outcomes. This innovative database architecture directly mitigates risks highlighted in CAA CAP 371 and Safety Directive 2023/05, reducing costly duty-hour violations, optimizing crew utilization, and enhancing fatigue management. Empirical evidence demonstrates substantial operational improvements—such as a notable 33% reduction in crew assignment violations at key airports and an 86% decrease in audit preparation efforts.

The implemented system also ensures high scalability, fault tolerance, and performance efficiency, positioning BritAir for sustainable future growth. Ultimately, the polyglot persistence model offers BritAir Connect a resilient technological foundation, driving enhanced passenger reliability, regulatory transparency, and operational agility within the increasingly complex UK aviation landscape.

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8. Appendix

Fig 10 – Code Snippet (violation type piechart)

```
# Categorize violations based on ActualHoursWorked
def determine_violation(hours):
    if hours > 12:
        return 'Exceeded Daily Hours'
    elif hours < 6:
        return 'Insufficient Duty'
    elif 10 < hours <= 12:
        return 'Approaching Limit'
    else:
        return 'Compliant'

crew_work_hours_df['ViolationType'] = crew_work_hours_df['ActualHoursWorked'].apply(determine

# Count each violation category
violation_summary = crew_work_hours_df['ViolationType'].value_counts()

# Plot the pie chart
plt.figure(figsize=(6, 6))
plt.pie(
    violation_summary,
    labels=violation_summary.index,
    autopct='%1.1f%%',
    startangle=140
)
```

Fig 11 – Code Snippet (Violation by Crew type)

```
# Aggregate total violations per CrewID
violation_counts = compliance_logs.aggregate([
    {"$group": {"_id": "$crew_id", "total_violations": {"$sum": 1}}}
])

violation_df = pd.DataFrame(list(violation_counts)).rename(columns={"_id": "CrewID"})

# --- Join SQL and MongoDB data on CrewID ---
merged_df = pd.merge(crew_df, violation_df, on="CrewID", how="left")
merged_df["total_violations"] = merged_df["total_violations"].fillna(0)

# --- Aggregate Violations by Role ---
role_summary = merged_df.groupby("Role")["total_violations"].agg(["sum"]).reset_index()

# --- Plotting ---
plt.figure(figsize=(7, 5))
plt.bar(role_summary["Role"], role_summary["sum"])
plt.title("Total Violations by Crew Role (SQL + MongoDB Union)")
plt.xlabel("Crew Role")
plt.ylabel("Total Violations")
plt.tight_layout()
plt.show()
```